Multiplication properties of exponents

1

a $3 \cdot v \cdot v \cdot u \cdot u \cdot u \cdot u \cdot u$

b $3 \cdot 2 \cdot m \cdot m \cdot m \cdot m \cdot n \cdot n$

2

a a^2b^2

b 3^4y^3

c $15u^3v^3$

d $-15c^4b^2a$

e $(3p)^3$

3

a a^3

b a^1

c $3x^8$

d x^2y^7

e x^6y^7

4

a 2^5

b 3^{19}

c y^8

d $10x^7$

e $12y^7$

f $24y^4$

g $40y^{16}$

h $64y^7$

i $54m^4$

j $35y^7$

k u^{11}

l $216y^{15}$

m $72y^{17}$

n $72y^7$

o $120y^{14}$

p $30y^{16}$

5

a $40u^7v^6$

b 0

c $56m^8n^5$

d $8064u^5v^5w^3$

6

a $504y^{24}$

b $-32y^{15}$

c $-60y^{20}$

d $72y^{14}$

e $-36b^8$

f $21y^6zx^3$

g $-30v^5w^5u^5$

h $-96y^{11}$

i $56p^5q^8$

7

a j^{10}

b c^{18}

c f^{48}

d w^{12}

e $243y^{13}$

f u^{20}

g $w^{36}v^{24}$

h k^{31}

i $24r^{58}$

j $x^{67}y^{15}$

k $160r^{39}s^4$

l $p^{45}q^{60}$

m p^{72}

n x^{60}

8 $9c^{12}$ square units

9 w^{50}

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10

a

i No **ii** No **iii** Yes **iv** Yes

v Yes

b

i No **ii** Yes

c $\left(r^2\right)^4 = r^8$

11 x^{10}

12

a x^{12} **b** $64y^{12}$ **c** $64y^{12}$ **d** $9y^{12}$

e $4x^4$ **f** $-243p^{10}$ **g** $16u^{20}v^{20}$ **h** $16r^8s^4$

13

a $\left(q^6\right)^2$ **b** $\left(3q^4\right)^3$ **c** $3p^4q^9$

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